National Institute of Technology, Kurukshetra
Department of Mathematics

Mathematics for Data Science(MAIC 209)

Time: 6:00 PM TO 6:50 PM | Date: Sept 24, 2025 | Total marks: 15

Attempt all questions

1. Using elementary row operations find for what values of a, the following system has i) no solution, ii) a unique solution and iii) infinite number of solutions.

$$x + y + z = 3$$
$$2x + 5y + 4z = a$$
$$3x + \left(a^2 - 8\right) z = 12$$

(5)

2. (a) State and prove Rank–nullity theorem.

   (b) Let $T : \mathbb{R}^3 \to \mathbb{R}^3$, be the linear transformation given by

$$T(x, y, z) = (4x + y, 27z, 0)$$

   Compute the rank and nullity of T.

(3+2=5)

3. Let T be the linear operator of $\mathbb{R}^3$ which is represented in the standard ordered basis by the matrix

$$A = \begin{bmatrix} -9 & 4 & 4 \\ -8 & 3 & 4 \\ -16 & 8 & 7 \end{bmatrix}$$

Check whether T is diagonalizable or not. If yes, find an invertible matrix P such that $P^{-1}AP = D$, where D is a diagonal matrix. Also find the minimal polynomial of T. (3+1+1=5)